

The Gateway Decomposition

$$\frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$$



$$x = \frac{p+A}{4}$$



$$4 \mid (p+A)$$

Divisibility

$$y = \frac{pN+d}{A}$$



$$A \mid (pN+d)$$

Residue

$$z = \frac{pNy}{d}$$



$$d \mid N^2$$

Gateway

where $N = (p+A)/4$, A prime, $A \equiv 3 \pmod{4}$